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JanSahayak

AI-Based Multilingual Chatbot for Welfare Scheme Awareness

Challenge 1.1 | Domain: AI & Intelligent Systems

Project goal: Build a low-bandwidth, multilingual assistant that helps rural users discover welfare schemes, understand required documents, and move toward application readiness through WhatsApp, SMS, IVR, or a web demo.

Refined submission report for the working prototype. This version removes non-professional footer text, strengthens product reasoning, improves safety framing, and keeps scheme recommendations grounded in official-source catalogue records.

3 Languages in prototype English, Hindi and Tamil support	10 Focused schemes High-impact central schemes	4-6 turns User flow Numeric, low-literacy friendly	4 Channels Web, WhatsApp, SMS, IVR
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1. Executive Summary

JanSahayak is a low-bandwidth welfare discovery assistant designed for rural users who may have limited literacy, unstable data connectivity, or only access to a keypad phone. The prototype compresses scheme discovery into a short question flow and returns a personalised shortlist with document checklists in the selected language.

The project deliberately avoids an uncontrolled free-form chatbot. Welfare eligibility is a high-trust domain; giving a confident but false answer can waste a user's time, money, and trust. Therefore, the prototype uses rules plus retrieval over a curated scheme catalogue first. Any future LLM layer must be constrained to retrieved scheme records and must cite the scheme source.

Why this product choice is right

Design decision	Reason	Risk avoided
Focused 10-scheme catalogue	Better depth, clearer eligibility, and easier validation than attempting all schemes immediately.	Shallow or inaccurate coverage of hundreds of schemes.
Menu-first conversation	Works across WhatsApp, SMS and IVR; reduces typing burden and bandwidth.	Drop-off caused by long free-form forms.
Source-grounded output	Each scheme card links back to a trusted official source.	Hallucinated benefits or fake eligibility.
Human fallback path	Routes uncertain cases to CSC, Gram Panchayat, ASHA/NGO worker or department.	Over-reliance on automation for edge cases.

Core deliverable coverage

Problem requirement	How this report/prototype satisfies it
At least two regional languages	Prototype supports Hindi and Tamil in addition to English.
8-10 high-impact schemes	Catalogue covers 10 schemes across farming, health, housing, fuel, employment, maternity, insurance and pension.
4-6 simple eligibility questions	Flow asks language, need, rural/urban status, demographic attributes, occupation/economic status, and key conditions.
Downloadable/SMS-able checklist	Backend generates compact document checklist output suitable for PDF, SMS or assisted-print mode.
Low bandwidth channels	Architecture includes web demo plus Twilio-ready WhatsApp, SMS and IVR endpoints.

2. Problem Framing and Target Users

The bottleneck is not the existence of schemes; it is last-mile discovery and application readiness. Rural users often face four blockers: they do not know which scheme applies, they do not understand portal language, they fear documentation mistakes, and they cannot rely on stable data connectivity.

Persona	Typical need	Conversation path	Likely output
Small farmer	Income support, crop-linked benefits, health protection	Need: farming/income -> rural -> occupation farmer -> land/bank/Aadhaar status	PM-KISAN, PMJJBY/PMSBY, AB-PMJAY if vulnerable household indicators match
Gig/unorganised worker	Insurance, pension, emergency safety net	Need: insurance/pension -> occupation worker -> age/bank status -> income/self-employed status	PM-SYM, APY, PMJJBY, PMSBY
Woman head-of-household	Cooking fuel, maternity support, housing, health	Need: LPG/maternity/housing -> gender/age -> LPG/household/pregnancy status	PMUY, PMMVY, PMAY-G, AB-PMJAY where applicable

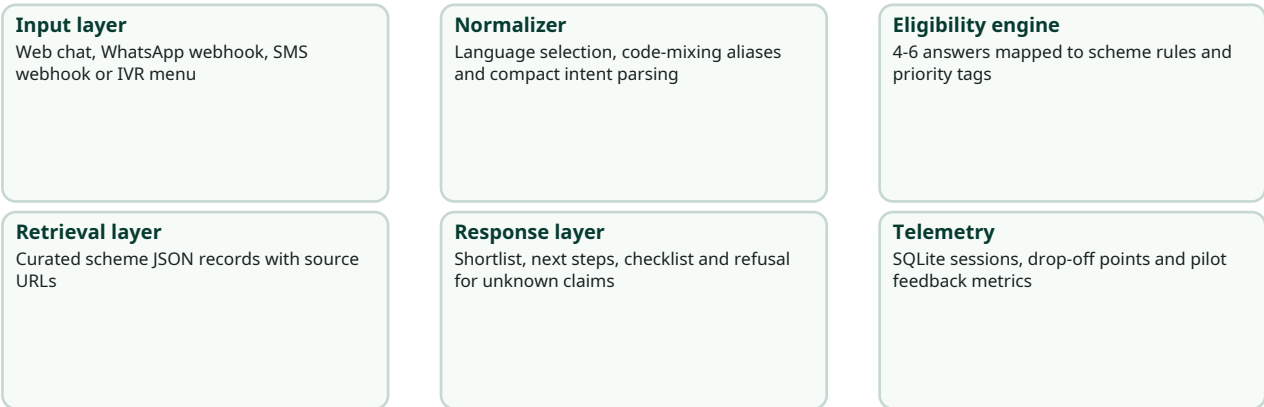
Lowest-friction channel decision

Channel	Strength	Weakness	Best use
WhatsApp	Familiar for smartphone users; supports buttons and PDF/checklist links.	Requires data and smartphone literacy.	Primary assisted-demo and semi-literate users.
SMS	Works on Rs. 500 keypad phones; no app install.	Very limited text length; language rendering can be inconsistent on old phones.	Fallback for checklist and numeric eligibility flow.
IVR	Works without literacy and without data; local-language voice menus.	More expensive and slower to build well; speech recognition for dialects is hard.	No-data/no-literacy fallback and assisted mode.
Web demo	Fast to test and present; good for judges and NGO workers.	Not the final rural channel by itself.	Prototype demonstration and field-worker console.

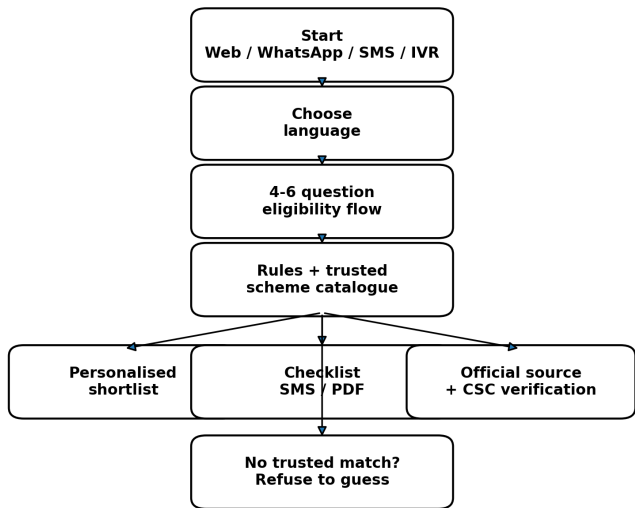
Ruthless assessment: WhatsApp is the best demo channel, but it is not the lowest-friction channel for the poorest users. The project is strongest when WhatsApp is paired with SMS/IVR and assisted CSC/NGO mode.

3. Product Architecture and User Flow

The system is built as a rules-and-retrieval chatbot with optional LLM extension. The safe default path is deterministic: collect minimal attributes, match against scheme rules, return a shortlist, and cite the source. This is intentionally less flashy but more defensible.



JanSahayak User Flow
Low-bandwidth welfare scheme discovery with hallucination guardrails



User flow diagram: short low-bandwidth interaction, source-grounded shortlist, and refusal-to-guess fallback.

Eligibility flow compressed into 4-6 turns

Turn	Question type	Example options
1	Language	English / Hindi / Tamil
2	Primary need	Farming, health, housing, LPG, work, maternity, insurance, pension
3	Location and household context	Rural/urban, household status, existing LPG/house condition
4	Age/gender/occupation	Adult woman, farmer, unorganised worker, pregnant/lactating, bank account holder
5-6	Key documents/conditions	Aadhaar, bank account, land record, ration card, job card, mobile number

4. Focused Scheme Catalogue

The prototype covers a focused catalogue of high-impact schemes. This is the right scope for a credible working demo: broad enough to cover common rural needs, narrow enough to keep eligibility explanations and document checklists auditable.

Scheme	Need area	Benefit summary	Checklist trigger
Pradhan Mantri Kisan Samman Nidhi (PM-KISAN)	farmer income support	Income support of Rs. 6,000 per year in three instalments to eligible landholding farmer families.	Aadhaar card, Land ownership/cultivation record, Bank account passbook
Ayushman Bharat - Pradhan Mantri Jan Arogya Yojana (AB-PMJAY)	health insurance	Cashless health cover for eligible poor and vulnerable families for secondary and tertiary hospitalisation.	Aadhaar card, Ration card/family ID where applicable, Mobile number
Pradhan Mantri Awaas Yojana - Gramin (PMAY-G)	rural housing	Assistance for eligible rural houseless households or households living in kutchha/dilapidated houses to build a pucca house with basic amenities.	Aadhaar card, Job card/ration card where applicable, Bank account details
Pradhan Mantri Ujjwala Yojana (PMUY)	clean cooking fuel	LPG connection support for eligible adult women from poor households without an existing LPG connection.	Aadhaar card, Proof of identity/address, Ration card or family composition document
Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)	rural employment	Demand-based wage employment for adult members of rural households willing to do unskilled manual work.	Job card application, Aadhaar/identity proof, Ration card/family details
Pradhan Mantri Matru Vandana Yojana (PMMVY)	maternal support	Maternity benefit support for eligible pregnant and lactating women under current PMMVY rules.	Aadhaar card, Mother and Child Protection/MCP card, Bank/post office account
Pradhan Mantri Jeevan Jyoti Bima Yojana (PMJJBY)	life insurance	Low-cost life insurance cover for eligible bank/post-office account holders in the specified age group.	Bank/post office account, Aadhaar/KYC, Mobile number
Pradhan Mantri Suraksha Bima Yojana (PMSBY)	accident insurance	Low-cost accident insurance cover for eligible bank/post-office account holders.	Bank/post office account, Aadhaar/KYC, Mobile number
Atal Pension Yojana (APY)	pension	Contributory pension scheme for eligible citizens, especially workers in the unorganised sector.	Bank account, Aadhaar/KYC, Mobile number
Pradhan Mantri Shram Yogi Maan-dhan (PM-SYM)	unorganised worker pension	Voluntary contributory pension for eligible unorganised workers.	Aadhaar card, Savings bank/Jan Dhan account with IFSC, Mobile number

Each shortlist item should be treated as a possible match, not a guarantee. Final approval still depends on official scheme rules, records, and verification through the relevant portal or local office.

5. Multilingual, Code-Mixing and Offline Design

The prototype supports English, Hindi and Tamil catalogue fields and keeps the conversation menu-based. Production deployment should add Bhashini/IndicTrans2 or an equivalent Indic NLP layer for stronger translation, speech support, and dialect coverage.

Challenge	Prototype handling	Production hardening
Code-mixing	Alias layer maps common phrases such as "mera aadhaar kho gaya", "gas nahi", "land job card", and Tamil farmer terms to normalized attributes.	Add ASR/TTS and translation pipeline; test with local dialects and actual user phrasing.
Low literacy	Numeric menus reduce typing and reading burden.	Add IVR voice prompts and field-worker assisted mode.
Old phones	SMS fallback uses compact replies and sends checklist in short text blocks.	Add short-code integration and local-language templates validated on feature phones.
No data balance	IVR/assisted mode routes user without internet.	Partner with CSC, Gram Panchayat, ASHA worker, NGO worker or district helpline.

Checklist generation

After a scheme match, the assistant generates a document checklist in the selected language. For SMS/IVR, it returns a compact checklist. For WhatsApp/web, it can provide a downloadable PDF or printable checklist for CSC/NGO-assisted application support.

Example: PM-KISAN checklist

Aadhaar card; land ownership/cultivation record; bank account passbook; mobile number; self-declaration/exclusion declaration where required.

Instruction: Visit the PM-KISAN portal or nearest CSC, complete registration, submit Aadhaar/land/bank details, and track beneficiary status.

6. Hallucination Control, Safety and Measurement

The most dangerous failure mode is a bot that sounds confident but gives wrong scheme eligibility. The project therefore treats scheme answers as source-grounded information cards, not final entitlement decisions.

Risk	Guardrail implemented	Remaining weakness
Hallucinated scheme answer	Unknown schemes trigger refusal to guess; official source URL is shown with every recommendation.	Static catalogue can become stale if not refreshed.
False eligibility confidence	Language uses "possible match" and "likely if"; never promises approval.	Users may still interpret shortlist as guaranteed entitlement.
Incorrect language translation	Menu-first prompts reduce long generated text.	Dialect and speech recognition need real field testing.
Sensitive data exposure	Prototype should not store Aadhaar numbers; telemetry logs only flow-level data.	Production needs consent, retention policy and access controls.
Exclusion of edge cases	Escalation to CSC/Gram Panchayat/department for final verification.	Human referral network must be operational.

Success metrics and instrumentation

Metric	Target	How to measure
Eligibility-flow completion	>= 80%	Session logs: users reaching shortlist/checklist divided by users who start language selection.
Comprehension score	>= 70%	Post-chat 3-question quiz: scheme name, benefit, next document/action.
Median response time	< 3 seconds	Backend timing under simulated 2G/3G latency.
Hallucination rate	Near zero	Manual audit: every recommendation must map to a catalogue record and source URL.
Checklist usefulness	>= 75% positive	Pilot users/field workers rate whether checklist is clear and actionable.

7. Pilot Plan, Impact Projection and Next Steps

The correct next step is not adding more AI. It is validating whether real users complete the flow, understand the answers, and can act on the checklist. A fake pilot would weaken the submission; the report should clearly separate prototype dry-runs from real field testing.

Pilot component	Minimum design
Sample	10-15 users from a nearby village/community; include at least one farmer, one woman head-of-household and one unorganised/gig worker.
Test method	Run each user through the chatbot in their preferred language/channel; record completion, questions asked, confusion points and checklist comprehension.
Post-chat quiz	Ask three simple questions: What scheme did you get? What benefit does it provide? Which documents will you carry?
Ethics	Do not collect Aadhaar numbers; take verbal consent; explain that final eligibility must be verified by official authorities.

Adoption scenario	Assumption	Projected value
District/NGO pilot	10,000 reachable beneficiaries	Can identify eligible users and route them to scheme-specific application support.
Conservative uptake lift	20% of users successfully act on at least one relevant scheme	2,000 additional households moved from awareness to application readiness.
PM-KISAN example	If 500 additional farming households successfully enroll at Rs. 6,000/year	Rs. 30 lakh/year entitlement flow for that one scheme alone.
Health protection example	Even a small number of AB-PMJAY card usages can prevent catastrophic medical debt	High social value beyond direct cash transfer calculations.

Build roadmap

Phase	Work to complete
Prototype hardening	Replace static data with scheduled official-source refresh; add state/district scheme filters.
Language hardening	Add Bhashini/IndicTrans2 pipeline and test with real code-mixed/dialect inputs.
Field validation	Run the 10-15 user pilot, update metrics, and document drop-off reasons.
Operational deployment	Create NGO/CSC dashboard, escalation paths, privacy policy and consent flow.

8. Official Source Grounding

The prototype catalogue uses official-source URLs for scheme grounding. The chatbot must never answer beyond the verified catalogue unless a fresh official source has been retrieved and logged.

Scheme	Official source used in catalogue
Pradhan Mantri Kisan Samman Nidhi (PM-KISAN)	myscheme.gov.in/schemes/pm-kisan
Ayushman Bharat - Pradhan Mantri Jan Arogya Yojana (AB-PMJAY)	myscheme.gov.in/schemes/ab-pmjay
Pradhan Mantri Awaas Yojana - Gramin (PMAY-G)	myscheme.gov.in/schemes/pmay-g
Pradhan Mantri Ujjwala Yojana (PMUY)	myscheme.gov.in/schemes/pmuy
Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)	myscheme.gov.in/schemes/mgnrega
Pradhan Mantri Matru Vandana Yojana (PMMVY)	myscheme.gov.in/schemes/pmmvy
Pradhan Mantri Jeevan Jyoti Bima Yojana (PMJJBY)	myscheme.gov.in/schemes/pmjjby
Pradhan Mantri Suraksha Bima Yojana (PMSBY)	myscheme.gov.in/schemes/pmsby
Atal Pension Yojana (APY)	myscheme.gov.in/schemes/apy
Pradhan Mantri Shram Yogi Maan-dhan (PM-SYM)	myscheme.gov.in/schemes/pm-sym

Final position: JanSahayak is strong enough for a hackathon or stakeholder demo. It should not be released as a production eligibility authority until source refresh, state-specific rules, privacy controls, and a real user pilot are completed.